Whether Chemical Warfare Agent Simulant Designations Look Supported Depends on Which Compounds Are in the Comparison

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Abstract

Chemical warfare agent detection is developed almost entirely on simulants: compounds accepted as agent-like and safe to handle. We tested whether that acceptance has a measurable basis. For eleven authentic agents we computed structural and GFN2-xTB electronic descriptors and asked, by exact permutation over every designation, whether three incumbent simulants sit closer to the agents than random compounds. On our primary thirty-five-compound panel, which includes five compounds the sensor literature also calls simulants, no agent has a valid match at any threshold and no test reaches significance. Resampling the descriptors gives the incumbent three a median of zero matched agents against five for the eight the field uses, in all 1000 replicates. Distances barely move between panels; what changes is that something closer is present. A twenty-eight-compound panel omitting those five yields three matches and a nominal p of 0.025 that no correction over the eighteen tests leaves significant. Moving two labels, so that two organophosphorus pesticides count as simulants, produces an adjusted value of 0.0006. Two of the three were designated for measurements that are not detection; the third was never designated. A defensible claim must name the originating measurement and fix panel, space, statistic and weighting in advance.

Keywords: chemical warfare agent simulant, descriptor space, DMMP, exact permutation test, multiplicity, acute exposure guideline level, dual-use

1. Introduction

Work on detecting chemical warfare agents (CWAs) is rarely done with CWAs. Outside a small number of licensed facilities, sensor arrays, sorbents and sampling protocols are developed, optimised and published using simulants: compounds chosen to behave like an agent while being safe enough for an ordinary laboratory. The choice of simulant therefore sets a ceiling on what a published detection result can be taken to mean, and that choice is made by a convention rather than by a measurement.

The dependence is structural, not incidental. The standing reviews of agent destruction and detection organise a literature in which reactivity and sensing chemistry are established on surrogates and extrapolated to agents [1, 2], and the same pattern runs through the chromogenic,

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fluorogenic and supramolecular recognition chemistry developed for nerve agents [3, 4, 5]. Much field instrumentation descends from a colorimetric sensor-array lineage established on ordinary analytes and odours [6, 7, 8] and transferred to the agent problem, so the compounds an array is characterised on are inherited rather than fixed by the chemistry. Our own work sits on both sides [9, 10, 11], and detection closed against authentic agents is rare enough that surrogate work is where most of the field's evidence is generated.

What a simulant is for differs by subfield, and so does what makes one adequate. For environmental fate and transport the criterion has been reviewed explicitly [12]; in metal–organic framework adsorption and catalytic hydrolysis the simulant establishes the mechanism and the kinetics [13, 14]; in decontamination the organophosphorus simulant dimethyl methylphosphonate (DMMP) is the compound whose removal is quantified [15]. In every case the inference runs from the simulant to the agent, and the warrant for it is a resemblance argument.

The convention is property matching. A recent critical review of liquid, low-toxicity simulants states the criterion plainly: align physicochemical properties such as hydrophobicity, volatility and solubility with those of the agent, with the simulants it assesses described as presenting minimal toxicity risks [16]. That review is confined to liquid-phase sampling, but the same criterion underwrites vapour-phase detection, where the quantity of interest is not partitioning but a chemical response.

Two assumptions are built into this practice, and both are testable. The first is that resemblance in structural and physicochemical descriptors implies resemblance in the property a sensor reports. For adsorption in metal–organic frameworks this has been examined directly: four common simulants and two nerve agents simulated across 2969 frameworks reproduce agent adsorption behaviour only partially, and the agreement depends on which framework and which agent [17]. The analogous question for the electronic properties governing a chemical response has not been asked. A cheminformatic attempt at the selection problem predates this one, in this journal: Lavoie et al. applied the chemical similarity search used in pharmaceutical screening to simulant identification, with the stated aim of finding compounds close to the agents but less toxic [18]. That aim is the one the field still has. What the present work adds is that the compounds the detection literature settled on were not reached by that procedure or by any comparable one we could find. The second assumption is that a simulant is safe, usually supported by the absence of an agent-class hazard label rather than by a toxicity value, although where simulants are handled at scale the protective equipment is specified against measured permeation data [19]. That is the practice in detection; it is not the practice everywhere. The human-decontamination literature screens candidates against toxicity explicitly and reports the attrition. One systematic review identified 22 simulants and judged 17 unsuitable for human volunteer trials [20]; a later one screened 171 chemicals, discounted 78 on the risk they would pose to volunteers, and assembled the remainder into a property-indexed repository of 39 suitable and 54 provisional compounds [21]. The two literatures select under different criteria because the measurements differ: a decontamination trial is governed by volatility and solubility, a sensor response by molecular recognition.

Both assumptions can be checked against measurements that already exist. The electronic structure of small molecules is accessible at semi-empirical tight-binding cost [22, 23, 24], and acute inhalation toxicity has been derived by a formal regulatory process for several hundred chemicals, including the agents themselves [25]. Ranking compounds along a sparse regulator-derived hazard axis bounded by an explicit applicability domain is the approach we have applied elsewhere [26]; here it is turned on the compounds the detection field uses on itself. We ask where simulants and agents sit relative to one another, how much of that answer is fixed by the data rather than by the analyst's choice of panel, space and statistic, whether the incumbents are safe

and whether that claim rests on a value or on an absence, and whether fidelity and safety are coupled.

The first question turned out to have a different shape than we expected. We began by comparing descriptor spaces, found a difference, and then found on audit that it did not survive extending the compound panel, correcting for the number of statistics computed, or removing a descriptor that was an exact function of two others in the same space. What the data support is reported below as such, with the earlier reading alongside it, because the way a nominal result of this kind arises is part of what a reader of this literature needs.

Dual-use statement.. Throughout, the optimisation direction is fixed: fidelity to the detection-relevant properties is to be maximised while phosphorus electrophilicity, acetylcholinesterase reactivity, alkylating capacity and predicted mammalian toxicity are to be minimised. No analysis here ranks compounds by potency. The structural filter in Section 2.7 excludes candidates and never orders them, and we did not extend the toxicity model of Supplementary Section S0.2 to the organophosphorus class, because a model predicting mammalian lethality for structures not yet made would invert the direction of the work. That a predictive model trained for safety can be run in reverse is not hypothetical [27], and how much of such a method to disclose is itself a live problem [28]. Mechanistic modelling of organophosphorus lethality is established and separately published, including by the present authors [29]; what is withheld is its use as a design objective over novel structures, not the literature's existence.

2. Methods

2.1. Compound set and the two panels

Eleven authentic agents are common to both panels: sarin GB, soman GD, tabun GA, cyclosarin GF, VX, sulfur mustard HD, lewisite 1 L, hydrogen cyanide AC, cyanogen chloride CK, phosgene CG and chloropicrin PS. Three compounds are treated as the incumbent simulants: DMMP, dichloromethane and dipropylene glycol monomethyl ether (DPGME).

We call these three incumbent rather than designated. Searching for an instrument designating them as chemical warfare agent simulants, we found that two carry a documented designation for a purpose other than the one they now serve, and the third carries none.

DMMP was identified as a test simulant for canister and cartridge filtration against GB and HD in a NIOSH respirator programme [30], a document that also states the simulant information it provides is not to be used as a recommendation of any simulant, and whose property at issue is adsorptive breakthrough rather than detection chemistry. DPGME appears as "a preferred G agent simulant" in a patent assigned to the United States Department of the Army [31], in a training composition formulated with a fluorescent dye and justified on ion-mobility detectability, so the application is field training rather than sensor validation. For dichloromethane we found no designating document of any kind: it is absent from the NIOSH candidate list and from the Man-in-Simulant Test programme, and present in this role, as far as we can establish, only in the research literature.

A negative claim about documents is only as good as its scope. We searched the standard-setting and test-procedure literature reachable without a subscription or a clearance: NIOSH and NPPTL respirator documents, US Army and DEVCOM reports indexed by the Defense Technical Information Center, United States patents assigned to the Department of the Army, National Academies assessments of Department of Defense test programmes, ASTM and NATO standards

catalogues, and the peer-reviewed simulant literature cited here. Two Test Operations Procedures bearing on the question (TOP 08-2-196A and TOP 10-2-022B) we could not obtain, and national CBRN defence manuals are generally not public. A designation in one of those would change the paragraphs above but not the measurements below.

Two observations follow. A designation is specific to a measurement: a compound chosen because it breaks through a carbon bed like sarin has been validated for adsorption and not for an optical or electrochemical response, and one chosen because it lights up an ion-mobility detector in a training lane has been validated for that detector. The sensor literature using these three is using designations issued for other purposes. The compounds with the clearest designations, in turn, are not these three: methyl salicylate is the simulant of the United States Army's Man-in-Simulant Test programme, chosen for sulfur mustard on 1994 penetration trials [32], and the NIOSH candidate list consists of DFP, DIMP, DEEP, TEP, dibutyl sulfide, CEES, malathion and methyl salicylate. Those compounds are in the second panel described below, not the first. A recent survey reaches the same conclusion in general form, that simulants have been selected largely on historical use and that no standard for chemical agent simulants exists [33].

The two panels differ only in the non-agent pool, and the difference is the point. The twenty-eight-compound panel adds fourteen interferents and solvents routinely reported in sensor-array selectivity panels. The thirty-five-compound panel adds, to those, five further compounds used as simulants in the sensor, sorbent and decontamination literature (diisopropyl fluorophosphate DFP, diethyl chlorophosphate DCP, diethyl cyanophosphonate DCNP, 2-chloroethyl ethyl sulfide CEES and methyl salicylate), classified as simulants rather than as interferents, together with two organophosphorus pesticides, dichlorvos and malathion. We class the pesticides as interferents because the sensor and decontamination literature we surveyed does not use them to stand in for an agent, and flag the decision as contestable on two counts: malathion is on the NIOSH respirator candidate list quoted above, and both pesticides carry the phosphoryl motif that the structural filter of Section 2.7 treats as agent-class. Section 3.1 reports how sensitive the result is to exactly this kind of decision.

The thirty-five-compound panel is the primary one, and the five it adds are the ones with documentary standing: DFP and methyl salicylate appear in the NIOSH candidate list quoted above, methyl salicylate is the Man-in-Simulant Test simulant [32], and all five are in current use. A comparison asking whether three compounds are close to the agents while excluding the obvious alternatives is not a test of those three. We report the twenty-eight-compound panel throughout with equal numerical detail, because what it shows beside the larger one is the paper's result, and not to support a conclusion. All class labels were read from a single canonical table and never assigned by hand.

Panel, space, statistic and descriptor weighting were not fixed before the analysis was run, and this paper's conclusion is that they should be. We state them here so that the standard we recommend can be applied to the work recommending it. The thirty-five-compound panel is primary because the five compounds it adds carry documentary standing the incumbent three do not. The merged space is reported alongside both component spaces throughout rather than selected among them. Six distance statistics were computed and none was nominated in advance, which Section 3.1 reports the consequence of and Table 2 prices. Equal weight per descriptor was the default we inherited rather than one we chose, and Supplementary Section S0.7 reports the alternative.

2.2. Descriptors

Structural and physicochemical descriptors (51 after excluding run metadata) were computed with RDKit [34] from canonical SMILES, and include the Crippen atomic-contribution estimates of logP and molar refractivity [35], the fragment-based topological polar surface area [36] and the κ shape indices [37]. Interpretable descriptors were used throughout in preference to a substructural fingerprint such as ECFP [38], because the quantity of interest here is the position of a compound along named physical axes rather than a similarity score. Identifiers, formulae and molecular weights were resolved against PubChem [39]; the EPA CompTox Dashboard [40] was the intended second route but its programmatic interface was not reachable from either network used here, so identities rest on PubChem and on the identifiers printed in the regulatory source documents, and no property value was taken from a prediction service.

Electronic descriptors were computed with GFN2-xTB [23], the second-generation parametrisation of the geometry–frequency–noncovalent tight-binding family [22, 24], in the gas phase; the implicit solvation model available for these methods [41] was not applied, since vapour-phase response is the property of interest. Conformers were generated with ETKDGv3 [42, 43], optimised with MMFF94 [44] (UFF [45] for lewisite 1, the only compound lacking MMFF parameters), then optimised and evaluated at GFN2 level. Ensembles were enumerated from the distance-geometry stage rather than through a dedicated sampling workflow [46], as these descriptors are not strongly conformation dependent. Properties for the 28-compound set were Boltzmann averaged over up to five conformers at 298.15 K with tight optimisation; the larger AEGL screening set used a single conformer with `-opt normal`. On the 28 compounds the two protocols agree to Pearson $r \geq 0.9857$ across the four electronic descriptors, with mean absolute differences of 0.029 to 0.052 eV for the orbital quantities and 0.11 D for the dipole, the largest single dipole discrepancy being 0.95 D at tabun. Which protocol an analysis uses is fixed by what it compares. The descriptor-space comparison of Section 3.1, which stays inside the 28 compounds, uses the Boltzmann-averaged values. The decoupling analysis of Supplementary Section S0.13 places the 28 and the AEGL pool in one space, so it uses the single-conformer values for all of them, including the 28; mixing protocols within a distance calculation would put part of the systematic difference above into the distances.

Four electronic descriptors are available for every compound: E_{HOMO}, E_{LUMO}, the global electrophilicity index $\omega = \mu^2/2\eta$ in the definition of Parr et al. [47], and the dipole moment. Four further descriptors exist only for the six phosphorus-bearing compounds: the Mulliken charges at phosphorus and at the phosphoryl oxygen, the Wiberg P=O bond order [48], and the condensed Fukui index f^+(P) for nucleophilic attack at that centre [49, 50, 51]. They are reported in Supplementary Section S0.9 and excluded from any analysis spanning all 28 compounds.

2.3. Descriptor spaces and distances

Three spaces were compared: structural, electronic and merged.

The electronic space carries three coordinates, not the four that frontier-orbital descriptor sets of this kind usually report. The fourth, the Parr electrophilicity index $\omega = \mu^2/2\eta$ with $\mu = (E_{\mathrm{HOMO}} + E_{\mathrm{LUMO}})/2$ and $\eta = (E_{\mathrm{LUMO}} - E_{\mathrm{HOMO}})/2$ [47], is an exact function of the two orbital energies that accompany it. We confirmed this numerically: across the panel $|\omega - \mu^2/2\eta|$ reaches at most 3.6×10^{-15} eV. The correlation filter described below does not remove it, because $r(E_{\mathrm{LUMO}}, \omega) = -0.899$ falls below the $|r| > 0.95$ threshold; a filter built on linear correlation will pass a variable determined exactly by functional dependence. Three principal components of the standardised four-variable set account for 97.6% of its variance, and removing ω leaves the

rank order of pairwise distances almost unchanged (Spearman 0.932). We therefore take E_{HOMO}, E_{LUMO} and the dipole moment as the electronic space, and report ω and the condensed Fukui index in Supplementary Table S2 without admitting them to any distance. The four reactivity descriptors localised at phosphorus are undefined for the twenty-two panel compounds that contain none, so they cannot enter a panel-wide distance either.

Within each space, descriptors with zero variance were removed, then one of each pair correlated at $|r| > 0.95$ was removed with preference for retaining a protected set (molar refractivity, logP, TPSA, hydrogen-bond acceptor count, four linear-solvation-energy surrogates and the electronic descriptors). The four surrogates are linear rescalings of quantities already in the set and are not Abraham solvation parameters; three of the four are removed by the correlation filter at $|r| = 1.00$ in every space. The attrition is the same on both panels: no structural descriptor has zero variance, the correlation filter removes 13 of the 51, and the 38 that remain are the structural coordinates used throughout. Remaining descriptors were standardised and distances taken as Euclidean. Two shape indices, κ_2 and κ_3, go negative for the smallest molecules (water, $\kappa_3 = -104.0$); repeating every analysis without them returns the permutation values of Section 2.4 to three decimal places (0.028, 0.275 and 0.025), so no result below rests on the pathological indices.

2.4. Nearest-neighbour comparison and permutation test

For each authentic agent we recorded the distance to the nearest incumbent simulant and to the nearest interferent. With fourteen interferents against three simulants, the minimum of fourteen draws is smaller than the minimum of three by construction, so a paired test of the two distances is confounded by group size. We therefore enumerated all ways of designating three of the non-agents as simulants, computed the same statistic for each, and report the rank of the actual designation as an exact p-value. That is $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The null expectation for "an interferent is closer" follows from the group sizes alone and is 9.63 of 11 on the primary panel and 9.06 on the smaller one.

The hypothesis under test is directional, since property matching is claimed to place simulants closer to the agents than arbitrary compounds would be. The reported p is therefore one-sided, the fraction of designations whose mean d(nearest simulant) − d(nearest interferent) is at least as negative as the actual one's. The choice is consequential and we give all three conventions: scored two-sided as distance from the null mean the same statistic gives 0.029, 0.482 and 0.025 for the structural, electronic and merged spaces, and at twice the smaller tail 0.056, 0.550 and 0.050, so whether the structural value sits inside a conventional threshold depends on a choice a reader may make differently. This is the first of the analyst choices catalogued in Section 3.1 and the only one with a defensible default, the hypothesis as its proponents write it being directional.

2.5. Match criterion

Nearest-neighbour rank alone is not sufficient: in a 38-dimensional space a "nearest" simulant at a distance of 11.8 is not a match. We therefore normalised each distance by the median nearest-neighbour distance of the same space and define a valid match as a normalised distance ≤ 2 together with a distance smaller than that to every compound outside the designated set. The two conditions are not equally fragile: the threshold is a convention, the second condition is not. The criterion is a statement about proximity in this descriptor space and nothing further. It does not certify that a simulant reproduces an agent's response on a given instrument, and a compound

that fails it may still suit a measurement governed by one descriptor rather than by their aggregate. On the primary panel the incumbent simulants fail the second condition for all eleven agents, so the count there is zero at every threshold, including an unbounded one. We report the count at 1.5, 1.75, 2.0, 2.5 and 3.0 in Section 3.3, including the one setting at which it changes.

2.6. Descriptor bootstrap

The conclusions of Section 3.1 turn on differences that are small in absolute terms, 0.12 typical nearest-neighbour separations between DMMP and diisopropyl fluorophosphate at sarin, and a reader is entitled to ask how much of that the descriptor choice carries. We resampled the 41 merged-space descriptors with replacement 1000 times (seed `0xC0DE`), restandardised, recomputed the distance matrix and the median nearest-neighbour separation within each replicate, and recounted the matched agents. Resampling features rather than compounds is the relevant perturbation, because the compound set is fixed by the question and the descriptor set is not. The same replicates serve both simulant definitions, so the difference is paired.

2.7. Structural filter

A SMARTS filter was written to exclude candidates carrying agent-class motifs, following the structural-alert tradition in which a substructure associated with a mechanism of toxicity serves as a categorical screen [52, 53]; the motifs here are agent-class rather than generic and the filter excludes only. Tier A covers a phosphoryl group bearing a good leaving group (fluoride, cyanide, thiolate or activated aryloxy), arsenicals, sulfur and nitrogen mustard backbones, and cyanide-releasing motifs. Tier B covers phosgene-derived acyl halides and chloroformates, halogenated oximes, α-halo carbonyls, and groups making a compound impractical rather than agent-like (halosilanes, sulfonyl halides, peroxy acids). Tier C records a simple phosphoryl ester without a leaving group and is noted rather than excluded. An earlier version that did not distinguish leaving groups flagged DMMP at the same tier as sarin and admitted potassium cyanide to the faithful-and-safe quadrant; both were corrected.

2.8. Reproducibility

All analyses are deterministic given the seed and the permutation test is exhaustive rather than sampled. Numerical work used NumPy 2.5.3 [54], SciPy 1.18.1 [55], pandas 3.0.5 [56] and scikit-learn 1.9.1, figures Matplotlib 3.11.2 [57], and descriptors RDKit 2026.03.6 [34]. Code, descriptor tables, parsed toxicity values with their sources and the figure scripts are in the repository cited under Data and code availability.

3. Results

3.1. What the comparison contains decides the answer

On the thirty-five-compound panel, no authentic agent has a valid match to an incumbent simulant. Figure 1a shows why, and the mechanism is not the one our earlier version reported. The distances from the agents to the incumbent simulants hardly move when the panel is extended: sarin sits at 1.14 typical nearest-neighbour separations from DMMP on the smaller panel and 1.19 on the larger, soman at 1.42 and 1.45, cyclosarin at 1.48 and 1.58. The compounds did not become less faithful. What changed is that diisopropyl fluorophosphate is present, at 1.07, 0.99 and 1.17 from the same three agents, and the match criterion requires a simulant to be nearer than every compound not designated one. The same holds across the panel, with 2-chloroethyl ethyl sulfide at

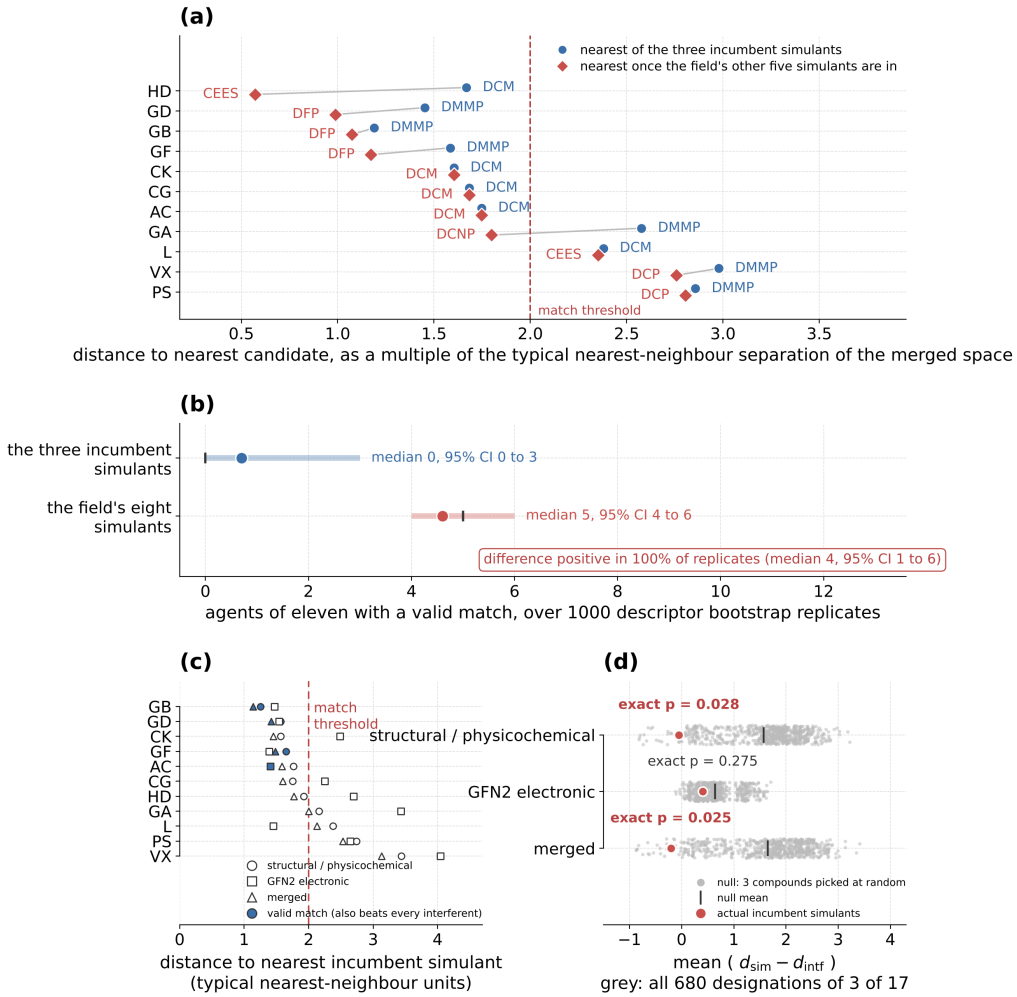


Figure 1: What the comparison contains decides the answer. (a) For each agent, the distance to the nearest incumbent simulant (circles) and to the nearest candidate once the five compounds the field also uses as simulants are included (diamonds), as multiples of the typical nearest-neighbour separation of the merged space on the thirty-five-compound panel. A match requires a normalised distance below the dashed threshold *and* a distance smaller than that to every compound not designated a simulant, so sarin, soman and cyclosarin fail on the primary panel despite lying inside the threshold. (b) Matched agents under 1000 descriptor bootstrap replicates (Section 2.6) for the incumbent three and for the eight the field uses; circles are means, vertical rules medians, bars 95% intervals. (c) On the twenty-eight-compound panel, which omits those five, the distance from each agent to its nearest incumbent simulant in each of the three spaces, normalised by the typical nearest-neighbour separation of that space; filled markers are valid matches. (d) Exact permutation null for that panel over all 680 ways of designating three of the seventeen non-agents as simulants, with the actual designation marked; the p shown is the mean-difference statistic only, the other five and the adjusted values being in Table 1, where the Benjamini-Hochberg value over its eighteen tests is 0.159.

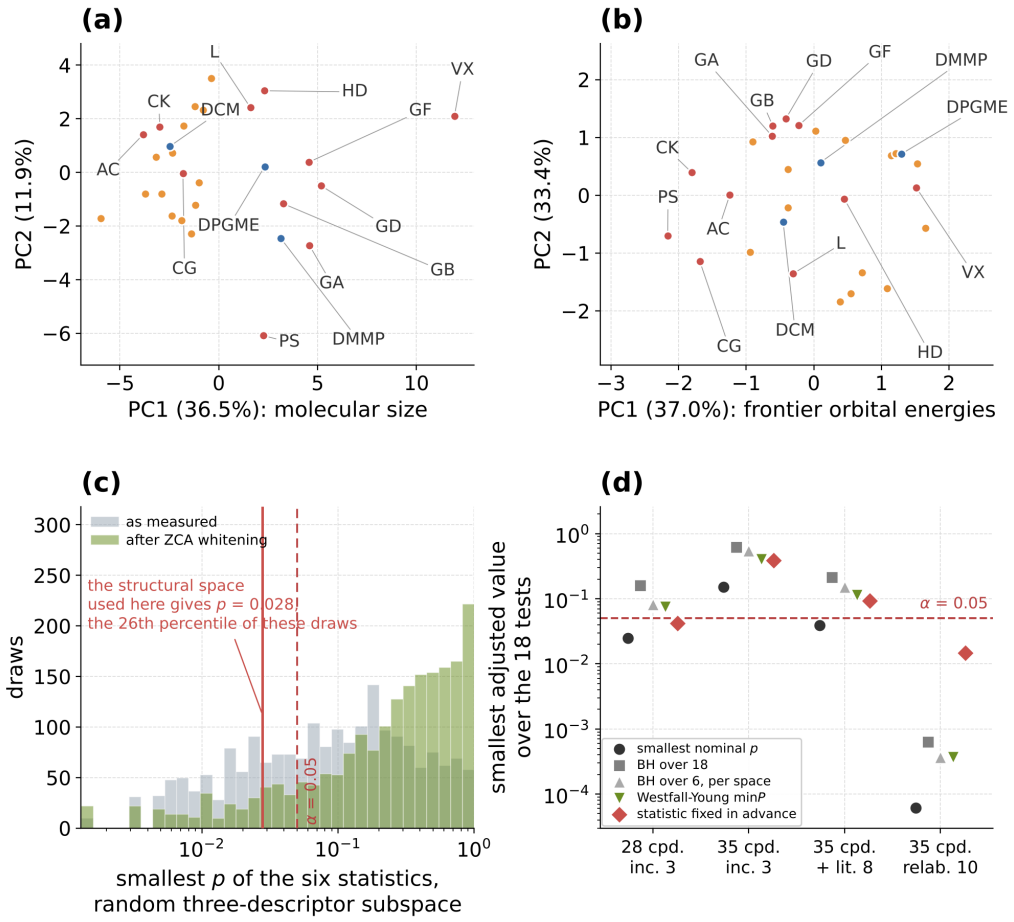


Figure 2: The spaces the comparison is made in, and what the nominal value is worth once the choices behind it are counted. Red marks the authentic agents, blue the three incumbent simulants and orange the interferents and solvents; only agents and simulants are labelled. (a) Principal components of the structural space and (b) of the three-coordinate GFN2-xTB electronic space; the leading structural component is a size axis and the leading electronic component loads the two orbital energies against the dipole moment. (c) Smallest p over the six statistics for 2000 random three-descriptor subspaces of the structural space, before and after ZCA whitening, with the space used here marked: its 0.028 is the 26th percentile of those draws, and 36% of them fall below 0.05 as measured, 17% after whitening. (d) Smallest adjusted value over the eighteen tests of each configuration under the five accounting schemes of Table 2; on the twenty-eight-compound panel only the scheme that fixes the statistic in advance falls below 0.05, and the fourth configuration falls below it under every scheme. Local reactivity at phosphorus and the rank of the Parr index are in Supplementary Figure S2.

0.57 from sulfur mustard where dichloromethane stands at 1.67, and diethyl cyanophosphonate at 1.80 from tabun where DMMP stands at 2.58. Admitting the five as simulants gives five matched agents of eleven, against three on the twenty-eight-compound panel and zero on the primary one. The zero does not depend on the threshold: no incumbent simulant is the nearest non-agent compound to any of the eleven agents, so the count stays at zero however far the threshold is relaxed.

That count is sensitive to the descriptors, and the honest statement of it is the interval rather than the point. Resampling the 41 merged-space descriptors with replacement (Section 2.6) gives the incumbent three a median of zero matched agents with a 95% interval of zero to three, and exactly zero in 58.8% of replicates, against a median of five and an interval of four to six for the eight compounds the field uses. The comparison is what survives: the difference between the counts is positive in 1000 of 1000 replicates, median four, interval one to six (Figure 1b). The per-agent reliabilities in the same replicates separate the firm results from the marginal ones. Sulfur mustard is matched by 2-chloroethyl ethyl sulfide in every replicate and tabun by diethyl cyanophosphonate in 52.5%, while under the incumbent set alone sarin recovers DMMP as its nearest compound in 37.8% of replicates, soman in 14.4% and cyclosarin in 12.7%. The displacement of DMMP at the G agents is therefore the weakest part of the picture and the sulfur mustard result the strongest, reversing what the point estimates suggest.

Figure 2a and b give the spaces. The leading structural component is a size axis carrying 36.5% of the variance and the leading electronic component loads on the two orbital energies against the dipole moment, the first two carrying 70.4%. Merging them returns a size axis at lower cumulative variance (45.9%) for a dimensional rather than a chemical reason: at equal per-descriptor weighting thirty-eight structural descriptors outvote three electronic ones.

Table 1 gives every exact permutation test we can run across four panel and designation configurations, three spaces and six statistics, in full rather than selected from, because the pattern across the cells is the finding.

Three things hold across the table. On the primary panel no test reaches nominal significance: in no space and under no statistic are the incumbent simulants distinguishable from compounds drawn at random. Classing the five literature compounds as simulants recovers two nominal values below 0.05, both in spaces containing the structural descriptors, so the difficulty is not that the test cannot detect anything on this panel. On the twenty-eight-compound panel, which differs only in omitting those five and two organophosphorus pesticides, the incumbents beat random compounds under two of six statistics in the structural space and three of six in the merged space.

Correcting for the number of times we looked removes almost all of it, and the exception is instructive. Table 2 gives the five correction schemes of Section 2.4. On the primary panel the question does not arise, since the smallest nominal value is already 0.151 under the per-descriptor standardisation used throughout. Weighting the 38 structural and 3 electronic descriptors as two equal blocks instead lowers that value to 0.076, which still survives no correction and leaves every valid-match count unchanged (Supplementary Section S0.7). The schemes matter only for the twenty-eight-compound panel. There the eighteen-test family gives a smallest adjusted value of 0.159, correcting the six statistics within each space gives 0.082, and the Westfall-Young min-P procedure, which uses the shared permutations and so accounts for the dependence between the tests exactly rather than approximating it, gives 0.075. That is the closest any correction brings the panel to the conventional threshold without crossing it. The Li-Ji approximation it replaces estimates 5.00 effective tests of eighteen and gives 0.125, conservative here by a factor near 1.7. None of the first three configurations of Table 1 reaches 0.05 under those four schemes.

The fifth scheme behaves differently. Had the mean difference been specified in advance as

Table 1: Every exact permutation test available. One-sided p for six distance statistics (Section 2.4) in three spaces, over four designation configurations on two panels (Section 2.1). q is the smallest Benjamini-Hochberg value over the eighteen tests of that configuration. Bold marks nominal $p < 0.05$. Only the fourth configuration, which relabels the two organophosphorus pesticides as simulants, reaches $q < 0.05$.

Panel / simulant set	Space	Dim.	$\Delta\bar{d}$	$\Delta\tilde{d}$	$\overline{\ln r}$	$\bar{d}_{\text{sim}}$	n_{closer}	n_{valid}	min q
35 cpd., incumbent 3	structural	38	0.326	0.343	0.404	0.237	1.000	1.000	
(primary)	electronic	3	0.386	0.374	0.412	0.344	0.723	0.664	
	merged	41	0.283	0.151	0.366	0.183	1.000	1.000	0.617
35 cpd., incumbent	structural	38	0.062	0.341	**0.050**	0.158	0.319	0.159	
+ literature 8	electronic	3	0.121	0.086	0.123	0.127	0.165	0.080	
	merged	41	**0.053**	0.326	**0.039**	0.128	0.319	0.159	0.212
28 cpd., incumbent 3	structural	38	**0.028**	0.176	**0.044**	0.085	0.110	0.216	
(omits the five)	electronic	3	0.275	0.175	0.293	0.235	0.757	0.757	
	merged	41	**0.025**	0.124	**0.032**	**0.041**	0.110	0.235	0.159
35 cpd., incumbent 3	structural	38	**0.013**	**<0.001**	0.055	0.251	0.318	0.268	
+ literature 5	electronic	3	0.089	0.053	0.059	0.086	0.066	0.055	
+ pesticides 2	merged	41	**0.005**	**<0.001**	**0.035**	0.166	0.318	0.268	**<0.001**

Statistics, in order: mean and median of $d_{\text{sim}} - d_{\text{intf}}$; mean $\ln(d_{\text{sim}}/d_{\text{intf}})$; mean d_{sim}; count of agents with a simulant closer than any interferent; count of valid matches. Permutations: $\binom{17}{3} = 680$, $\binom{24}{3} = 2024$, $\binom{24}{8} = 735\,471$, $\binom{24}{10} = 1\,961\,256$, enumerated exhaustively in each case.

the single statistic, three tests would have been run and the twenty-eight-compound structural and merged results would survive at $q = 0.042$, while neither thirty-five-compound configuration would (0.386 and 0.092). We did not specify a statistic in advance; we computed six and chose among them after seeing them, which is the circumstance the correction is for. The comparison is the paper's own argument made with its own data: one analysis, one data set, five defensible ways of accounting for the search, and a result significant under one and not the other four. The declaration that would have made the difference is one nobody in this literature makes, and it has to be made before the data are seen.

One label decision carries more than any of this. Section 2.1 classes dichlorvos and malathion as interferents and records that decision as contestable. Classing them as simulants instead, which moves two labels and changes nothing else, gives a smallest nominal value of 0.00006 and a Benjamini-Hochberg value of 0.0006 over the same eighteen tests, and it is the only configuration in this work that survives every one of the five accounting schemes of Table 2. The direction is not surprising, since both pesticides carry the phosphoryl motif the agents carry, and that is the point: significance here is available to an analyst who moves two labels, and the practice records neither which labels were moved nor that any were. This is not a recommendation. Neither pesticide is used as a simulant by anyone we can cite, and what the comparison shows is that a designated set can be made to look supported by choosing what goes into it.

One further observation bounds what the nominal values can mean. The count of agents for which a compound outside the designated set is nearer than any incumbent simulant is 11 of 11 in the merged space of the primary panel against a null expectation of 9.63 set by the 21:3 group-size asymmetry alone, and 6, 10 and 6 of 11 in the three spaces of the smaller panel against 9.06, so the raw count carries no evidence without the permutation test and in one space runs the wrong way.

Table 2: Five ways of accounting for the search, applied to the eighteen tests of each panel configuration. The Westfall-Young min-P value [58] is exact for the dependence among the tests, since all eighteen share one enumeration of designations; m_{eff} approximates that dependence from the eigenvalues of the correlation matrix of the eighteen nulls [59], used as a Bonferroni divisor. On the primary panel no scheme returns anything below 0.05 and on the smaller panel only the pre-specified-statistic scheme does, while the fourth configuration, which differs from the second by two class labels, is below 0.05 under every scheme.

Accounting scheme	35 cpd., incumbent 3 (primary)	35 cpd., + literature 8	28 cpd., incumbent 3 (omits the five)	35 cpd., relabelled 10 (pesticides in)
Smallest nominal p (18 tests)	0.151	0.039	0.025	<**0.001**
Benjamini-Hochberg over 18	0.617	0.212	0.159	<**0.001**
Benjamini-Hochberg over 6, per space	0.548	0.153	0.082	<**0.001**
Westfall-Young min-P over 18	0.402	0.113	0.075	<**0.001**
m_{eff} of 18 (Li-Ji)	6.00	5.00	5.00	5.00
Bonferroni at m_{eff}	0.907	0.196	0.125	<**0.001**
Statistic specified in advance (3 tests)	0.386	0.092	**0.042**	**0.015**

3.2. The difference between the spaces is not dimensional, and the nominal value is not informative

The structural space has 38 coordinates and the electronic space has 3, so a difference between them could be geometric rather than chemical. It is not. Truncating the standardised structural space to its first three principal components strengthens the result rather than weakening it, from $p = 0.028$ to $p = 0.0044$ on the twenty-eight-compound panel, which leaves the two spaces 62-fold apart in p at matched dimension. The second control is the one that matters. Of 2000 random three-descriptor subsets of the structural space, 36.3% reach $p < 0.05$, and the observed 38-dimensional value of 0.028 sits at the 25.8th percentile of that distribution while the electronic value of 0.275 sits at the 77.9th. Both are ordinary draws. The rate itself is inflated by correlation among the descriptors, falling to 16.9% on whitened coordinates, but the argument rests on the percentile and that is unaffected. Both controls are reported in full in Supplementary Section S0.6. The spaces do differ, and the difference is chemical rather than geometric; what the data do not support is that the difference says anything about the compounds that were designated.

3.3. How many agents have a valid simulant depends on the panel

Table 3 lists, for each agent on the twenty-eight-compound panel, the nearest incumbent simulant in the merged space as a multiple of that space's typical nearest-neighbour separation. Sarin (1.140), soman (1.417) and cyclosarin (1.483) are matched, all three by DMMP; tabun at 2.003 and VX at 3.131 are not. DPGME is nominally VX's nearest simulant in all three spaces, but at 3.131 typical separations in the merged space and 3.439 in the structural space that is not a match in any useful sense. Dichloromethane, used as a pulmonary-agent simulant, is not phosgene's nearest neighbour in any of the three spaces: acetone leads it in the structural space (1.14 typical separations against 1.75), ethyl acetate in the merged space (1.12 against 1.60) and chloroform in the electronic space (1.17 against 2.25).

The direction of the comparison matters. Table 3 gives, for each agent, its nearest simulant; the reverse question is which agent each simulant is nearest to, and over the same 1000 descriptor bootstrap replicates of Section 2.6 the three incumbents separate. DMMP is nearest sarin in 93.3% of replicates and nearest one of the four G-series agents in 99.0%. Dichloromethane is nearest a G-series agent in 1.4%, sitting instead with the choking and blood agents. DPGME divides evenly,

sulfur mustard in 47.0% and the G series in 46.7%. Being nearest is not being close, and DPGME carries no valid match in any space on this panel; but it is the one of the three whose designating document names a class, "a preferred G agent simulant" [31], and it is the one whose class these spaces do not resolve.

Table 3: Nearest incumbent simulant for each agent on the twenty-eight-compound panel, as a multiple of the typical nearest-neighbour distance of the merged space. A match requires a normalised distance ≤ 2 together with a distance smaller than that to every compound outside the designated set. On the primary thirty-five-compound panel no agent has a match.

Authentic agent	Nearest simulant	Normalised distance	Valid match
GB	DMMP	1.140	yes
GD	DMMP	1.417	yes
CK	DCM	1.451	no
GF	DMMP	1.483	yes
AC	DCM	1.585	no
CG	DCM	1.600	no
HD	DCM	1.768	no
GA	DMMP	2.003	no
L	DCM	2.128	no
PS	DMMP	2.530	no
VX	DPGME	3.131	no

The count is stable but not invariant. At thresholds of 1.5, 1.75 and 2.0 the merged space returns {GB, GD, GF} and at 2.5 and 3.0 it returns four, tabun being admitted once the bar passes its value of 2.003; the structural space behaves identically above 1.5 and at 1.5 narrows to {GB} alone, soman at 1.567 and cyclosarin at 1.648 falling outside. The three-agent figure is therefore a statement about the merged space at thresholds of 2.0 and below. The electronic space is the exception, and it is the informative one: there the count is one at every threshold tested, hydrogen cyanide matched by dichloromethane at 1.407, because acetonitrile sits closer to sarin, soman and cyclosarin than DMMP does and all three fail the threshold-free condition outright. Over the thresholds tested the count does not exceed four of eleven in any space.

The dependence on the panel is sharper than the dependence on the threshold. On the thirty-five-compound panel, with the incumbent three still the only simulants and the five literature compounds in the comparison pool, the merged space returns no valid match for any of the eleven agents: DMMP loses GB, GD and GF to diisopropyl fluorophosphate, which is nearer to all three. Admitting the five as simulants gives five matches of eleven, more than the incumbent set achieves on either panel, with DFP taking sarin, soman and cyclosarin at 1.07, 0.99 and 1.17, diethyl cyanophosphonate taking tabun at 1.80, and 2-chloroethyl ethyl sulfide taking sulfur mustard at 0.57, the closest agent-simulant pair anywhere in this work. VX remains unmatched in every panel and space we examined.

We draw no conclusion from this that the field should switch compounds, because the five were not selected by any procedure we can inspect either, and Section 3.1 applies to them exactly as it applies to the incumbent three. What the comparison establishes is narrower: the statement "three of eleven agents have a valid simulant" is true of one panel, false of a second, and replaced by a different list on a third, which is the same instability the permutation tests report and not an independent finding.

3.4. What the array actually measures

Every argument above rests on descriptor distances, and the premise that one of those spaces corresponds to what a vapour-phase sensor reports is tested in Supplementary Section S0.10 against measured 29-dye colorimetric response for twelve panel compounds. It does not resolve the question either way. Of eight panel-channel-space cells, two reach a nominal $p < 0.05$ under Mantel permutation over compounds and they favour different spaces; corrected across the eight, the smallest adjusted value is 0.098 and no cell reaches $q < 0.05$. With six compounds per plate set the design detects $|\rho| \geq 0.669$ at 80% power, so these data exclude a strong association and nothing weaker. Two artifacts are reported there because our own first version contained both: dye-axis centring fixes the between-compound mean cosine at $-1/(n-1)$, which the observed means clear by only 0.013 to 0.066, and the measured agent nearest to DMMP in the ultraviolet channel is tabun rather than sarin, which the descriptor criterion of Section 3.3 excludes.

3.5. The toxicity half of the criterion

The criterion has a second half: that compounds chosen on physicochemical grounds present minimal toxicity risk [16]. We tested it on the same panel and report it in full in the Supplementary Material. Three results are needed here.

Dichloromethane has a regulator-derived acute inhalation value and DMMP has neither an acute one nor a chronic one: it is absent from the AEGL list, and EPA's provisional assessment separately declined to derive a chronic inhalation reference concentration for want of data (Figure 3a and b, Supplementary Section S0.1). The two incumbents therefore differ in how well documented they are and not only in how hazardous, which matters because DMMP is also the compound against which decontamination efficiency is quantified [15]. Dichloromethane's value is favourable on the acute axis, at the 96th percentile of a 207-compound pool, and says nothing about the chronic and carcinogenic classification now displacing it from solvent inventories.

A model cannot fill the gap. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ and then predicts every one of twenty held-out military agents safer than measured, by factors of 2.5 to 2600, while certifying fourteen of them as inside its applicability domain (Figure 3c and d, Supplementary Section S0.2). The failure is not a missing mechanism, since thirteen training compounds carry an acetylcholinesterase-inhibiting pharmacophore and one is more acutely toxic than sarin. It is tail regression, a bias in the response space that an input-space domain check cannot see.

Fidelity and acute toxicity show no monotonic association across the 207 compounds carrying both measured axes (Spearman $r = +0.004$, 95% CI -0.132 to $+0.143$). The interval excludes a trade-off strong enough to constrain selection, which is weaker than independence and is what the data support, so the two can be improved separately (Supplementary Section S0.13). We draw no candidate recommendation from that: one resting on a single panel and two axes would repeat the error this paper reports.

4. Discussion

The criterion by which low-toxicity simulants are chosen is stated clearly enough to test [16]: align hydrophobicity, volatility and solubility with the agent. What the practice does not state is what the chosen compounds should be compared against, and that decides the answer. On a panel containing the compounds the field itself uses as simulants, the three incumbents match no agent

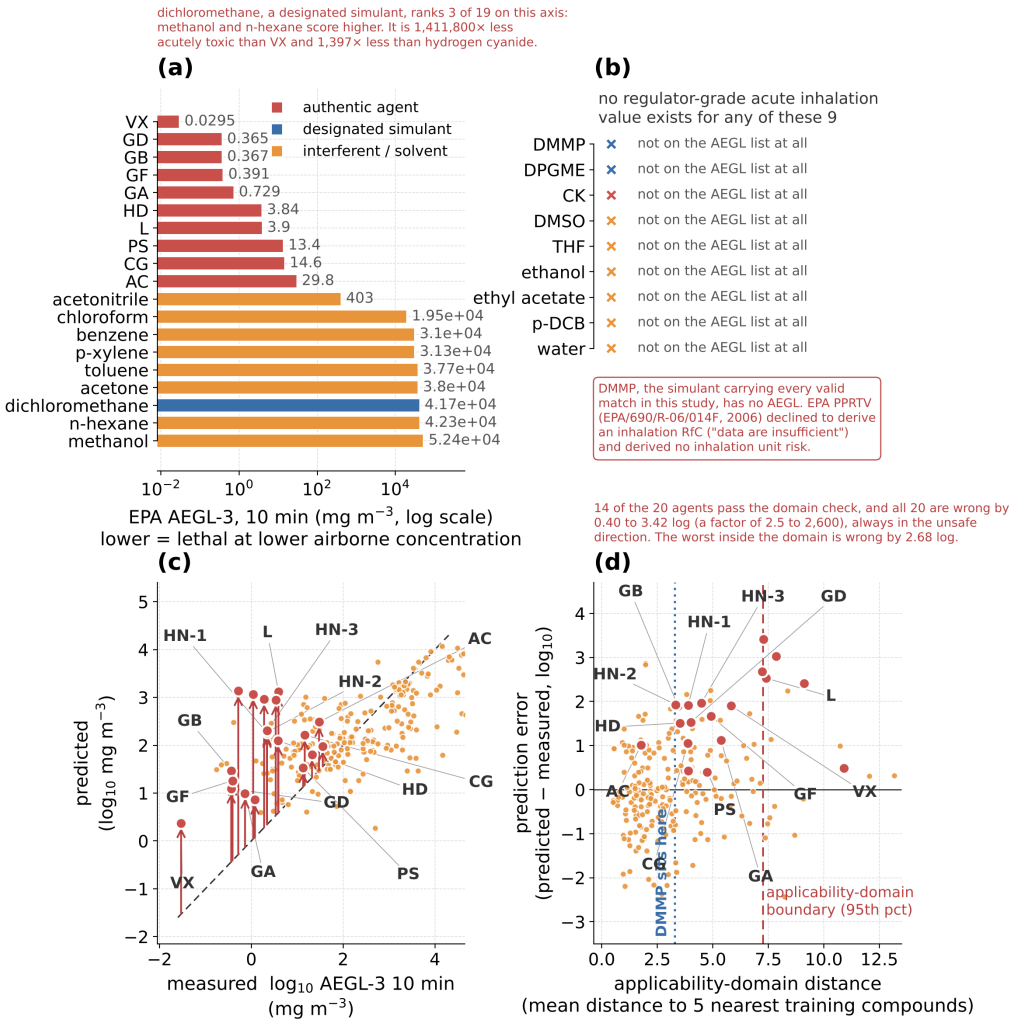


Figure 3: The second half of the criterion, measured and then modelled. (a) EPA AEGL-3 at 10 min on a logarithmic scale for the nineteen compounds that have one; the incumbent simulant dichloromethane sits near the safe end but is not the safest, methanol and n-hexane both scoring higher. Five of the nineteen values come from the agency's own footnotes rather than from the table cell, because they exceed half the lower explosive limit. (b) The nine compounds with no regulator-grade acute inhalation value, with the reason for each; DMMP, which supplies every valid match in this study, is among them. (c) Out-of-fold predictions for the 230 training chemicals (orange) with the twenty military agents held out of training (red) overlaid; arrows run from the measured value to the prediction, and every agent is predicted safer than it is. Labels name the ten agents of this paper's panel that the AEGL set also holds and the three nitrogen mustards; the remaining seven held-out agents are plotted unlabelled. (d) Prediction error against applicability-domain distance. Every one of the twenty is wrong by between 0.40 and 3.42 log units, and fourteen of them pass the domain check, the worst of those fourteen being wrong by 2.68 log units. The dotted line marks where DMMP sits.

and no test reaches nominal significance; on a panel omitting them, three agents are matched and the smallest nominal value is 0.025.

The mechanism is not a story about fidelity. The incumbents sit almost as close to the agents on one panel as on the other, moving by 0.03 to 0.10 typical separations (Section 3.1). What the larger panel supplies is diisopropyl fluorophosphate, nearer to sarin, soman and cyclosarin than DMMP is, so a criterion that asks a simulant to beat the alternatives returns nothing. Resampling the descriptors leaves that comparison intact in every replicate while separating the firm displacements from the marginal ones. A designation defended by a comparison that excludes the alternatives is not defended.

An earlier version of this work reported the twenty-eight-compound structural cell, $p = 0.028$, as the finding and built a thesis on the contrast between it and the electronic space. Three things undo that reading, and each is a mistake another group could make the same way. The value does not survive the eighteen tests of its own family (Benjamini-Hochberg $q = 0.159$), although it would have survived at $q = 0.042$ had the statistic been fixed in advance, which demonstrates the paper's argument rather than defending the value. It sits at the 25.8th percentile of 2000 arbitrary three-descriptor slices of the same block. And it is specific to the panel that omits the five.

Two things we expected to dissolve the contrast between the spaces did not. The Parr electrophilicity index was an exact function of two of the other electronic descriptors, so the electronic space never had the dimension we reported; removing it moved that space's value only from 0.297 to 0.275. Matching the dimensions did not dissolve the contrast either (Section 3.2). The contrast is real; what fails is the inference from it, because the structural value anchoring it is an ordinary draw from the subspaces we could have chosen.

What remains is a claim about comparison rather than about chemistry. On the primary panel we cannot establish that the incumbent simulants are closer to the agents than compounds drawn at random, and we cannot establish the opposite either. What we can establish is that the compounds which would make the comparison informative were not in it.

The documentary record explains how that arose, and it is not a story about carelessness. Two of the three were designated carefully for measurements that are not the one they are now asked to support (Section 2.1): DMMP for adsorptive breakthrough on a filter bed [30], DPGME for ion-mobility detectability in a training composition [31]. Each selection was fit for its own measurement. The compounds then carried their status into sensor development, where the governing property is the response of a dye or an electrode rather than a breakthrough time, and nothing in the original validation speaks to that. For dichloromethane we found no designating document at all.

The absence this paper documents is specific rather than general. A safety-screened repository of candidate simulants exists [21], and a descriptor-based selection procedure appeared in this journal fifteen years ago [18]. What is missing is any route by which either reached the compounds the detection literature settled on.

That mechanism is general. A simulant is validated against a measurement rather than against an agent, and treating the validation as a property of the compound, so that it travels into a new application, produces a working set whose standing no descriptor space can reproduce. The corrective is not a replacement compound but a declaration: which measurement a simulant was validated against, and a panel, space and statistic fixed before a claim that it is valid for a different one is tested. Agrawal et al. [17] reach the same place from the other direction, adequacy being a property of the compound together with the measurement it stands in for.

Three consequences follow for practice. DMMP is a surrogate for one G agent at a time rather than for the G series, and not a V-agent surrogate at all: the descriptor criterion assigns GB, GD

and GF to it, while the measured array puts sarin first in reflectance and tabun, which fails that criterion, first in the ultraviolet channel (Supplementary Section S0.10), and a sensor calibrated on DMMP meets a phosphorus centre 2.7 times more electrophilic than VX presents (Supplementary Section S0.9). The global electrophilicity index should not enter a descriptor space, for an arithmetic rather than a chemical reason: where the frontier orbital energies are present it adds no information, and a correlation filter will not notice. And the safety of a simulant is a question about evidence before it is one about hazard, since dichloromethane carries a regulator-derived acute inhalation value and DMMP carries none (Section 3.5).

We did not attempt to improve the toxicity model by training on organophosphorus compounds, and the distinction is narrower than a refusal to model the chemistry. Mechanistic structure–activity models of organophosphorus lethality exist and we have contributed to them [29]; such a model assesses compounds that already exist. What we decline to build is the same instrument turned into a design objective, a predictor of mammalian lethality evaluated over structures not yet made inside a loop that proposes them, which is the configuration in which a safety model becomes a potency model [27]. The cost is an acknowledged gap: for the compound class of greatest interest we can say that a predicted value is unreliable and cannot supply a better one. Reporting the gap rather than closing it is the correct outcome, and where to set that boundary is contested rather than settled [28].

Limitations.. The documentary findings of Section 2.1 rest on public sources, and the scope of that search is stated there. The claim is that no public instrument designates these three compounds for the use the sensor literature makes of them, not that no instrument does. A null result invites the question whether the design could have seen an effect. On the primary panel the mean-difference statistic reaches $p < 0.05$ once the incumbent simulants are moved 0.22 typical nearest-neighbour separations closer to the agents, while the deficit they actually carry has a median of 0.463 and falls below that value only at sarin, where it is 0.117 (Supplementary Section S0.8). The test is therefore sensitive to effects smaller than the one at issue. The agent set is eleven compounds and the nearest-neighbour statistics inherit that: on the larger panel with eight simulants, leaving out one agent at a time moves the structural p between 0.017 and 0.170, the worst case being removal of tabun, so a result one of eleven agents can carry across the 0.05 line is not a property of the panel. The permutation test is directional and reported one-sided, with both two-sided conventions given in Section 2.4. The two panels are the two we could build from compounds carrying complete descriptor data; measured toxicity values exist for a subset only, DMMP among the compounds without one, so the two axes do not run over the same compounds. A fourth configuration, classing dichlorvos and malathion as simulants, is reported in Table 1 and Section 3.1 because we flagged that label decision as contestable; the four configurations sample the configuration space rather than census it. The measured-array intervals describe variation across concentrations within one plate set rather than agreement between preparations (Supplementary Section S0.10). GFN2-xTB is semi-empirical, and the quantities used here are relative within a consistent protocol rather than absolute.

5. Conclusions

Simulant validity is not a property of a compound. It is a property of a compound set against a comparison, and the practice of choosing chemical warfare agent simulants does not record what the comparison should contain. On a panel of thirty-five compounds that includes the five the sensor, sorbent and decontamination literature also uses as simulants, not one of eleven authentic

agents has a valid match to an incumbent simulant and not one of eighteen tests reaches nominal significance. The incumbents are no further from the agents on that panel than on the smaller one; diisopropyl fluorophosphate is simply nearer, and under descriptor resampling the incumbent three match a median of zero agents against five for the compounds the field uses, the difference positive in every one of 1000 replicates. Omitting those five restores three matches and a nominal structural p of 0.028, a value at the 25.8th percentile of 2000 arbitrary three-descriptor subspaces of the same block which no correction over the eighteen tests run leaves significant.

The documentary record gives the reason. DMMP was designated for filtration breakthrough testing and DPGME for ion-mobility training compositions, neither of which is detection chemistry; for dichloromethane we found no designating instrument at all. A simulant is validated against a measurement, and that validation does not travel with the compound into a new application.

The absence of evidence is not evidence of unsuitability, and we draw no conclusion that the incumbents are poor choices. What we conclude is narrower and addressed to method. A simulant claim is testable, this field has the data to test it, and the test returns whatever the analyst's undeclared choices imply. Moving two labels, from interferent to simulant, is enough to turn a result that survives no correction into one with an adjusted value of 0.0006. The practical form of that conclusion is a reporting rule: a simulant result should name the measurement the compound was validated against and the comparison set it was tested in, and should fix the panel, the descriptor space, the statistic and the descriptor weighting before the test is run. Until those are declared in advance, a statement that a compound is a validated simulant reports a convention rather than a measurement.

Findings from the toxicity axis are in the Supplementary Material and do not depend on the contested comparison. The two incumbents differ in how well documented they are and not only in how hazardous, dichloromethane carrying a regulator-derived acute inhalation value and DMMP carrying none of any kind (Section 3.5). A random forest trained on 230 AEGL chemicals predicts all twenty held-out military agents safer than measured, by factors of 2.5 to 2600, so predicted toxicity cannot stand in for the missing values (Supplementary Section S0.12). Fidelity and acute toxicity show no monotonic association across the 207 compounds carrying both axes (Supplementary Section S0.13). We propose no replacement compound, because proposing one would repeat the error this paper is about.

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Ku Kang: conceptualisation, methodology, software, formal analysis, investigation, data curation, visualisation, writing – original draft. **Yoon Jeong Jang:** validation, writing – review and editing, supervision, project administration.

Declaration of competing interest

The authors declare no competing financial interest.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used a large language model (Anthropic Claude) to assist in drafting and editing prose, in writing and auditing the analysis and figure scripts, and in building the internal-consistency checks described under Reproducibility. No numerical result was produced by the model: every value reported here is read from the output of the deposited scripts, which were executed to generate it. The authors reviewed and edited all content and take full responsibility for it.

Data and code availability

Descriptor tables, parsed AEGL values with their sources, permutation distributions, candidate tables and all analysis and figure scripts are in a public repository at `https://github.com/bisu9082/simulant-designation-evidence`, at the tagged release `v1.0`, and are also provided with this article. The repository also carries the primary EPA source document from which every toxicity value here was parsed, so the parse can be checked against the original. The exact permutation test is exhaustive and the remaining analyses are deterministic given the stated seed.

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